- In February 2022, Russia launched a military offensive against Ukraine.
- On 24 February 2022 Ukraine informed the International Atomic Energy Agency (IAEA) that Russian forces had taken control of all facilities of the Chernobyl nuclear power plant. Control of the site was returned to Ukrainian personnel on 31 March 2022.
- In the early hours of 4 March 2022 the Zaporizhzhia plant in southeastern Ukraine became the first operating civil nuclear power plant to come under

Privacy - Terms

armed attack. Fighting between forces overnight resulted in a projectile hitting a training building within the site of the six-unit plant. Russian forces then took control of the plant. The six reactors were not affected and there was no release of radioactive material.

- Since late October 2022, Russia has repeatedly targeted Ukraine's civilian infrastructure, including the country's energy system, with missile strikes. Widespread blackouts have resulted, and external power supply to all four of the country's nuclear plants has been affected.
- Since January 2023 the IAEA has had a permanent presence at all of Ukraine's nuclear power plants.
- On 14 February 2025 a drone struck the roof of Chernobyl's New Safe Confinement (NSC). The IAEA said that both the outer and inner cladding of the NSC's arch had been breached, but that radiation levels were stable.

The International Atomic Energy Agency (IAEA) is closely monitoring developments in the country with respect to its nuclear facilities and is providing regular updates on the situation.

Timeline – key events

2022

24 February

Ukraine disconnected its grid from Belarus and Russia, and requested emergency synchronization to the European power grid.

Ukraine informed the IAEA that Russian forces had taken control of all facilities of the Chernobyl nuclear plant.

2 March

Russia informed the IAEA that its military forces had taken control of the territory around Ukraine's Zaporizhzhia nuclear plant. It said that radiation levels remained normal and added that personnel at the plant continued their "work on providing nuclear safety and monitoring radiation in normal mode of operation."

3 March

The State Nuclear Regulatory Inspectorate of Ukraine (SNRIU) reported that a large number of Russian tanks and infantry "broke through the block-post" to the town of Enerhodar, a few kilometres from the Zaporizhzhia nuclear plant. Power from one of the two offsite power transmission lines supplying electricity to the site was lost overnight. The power line does not provide power to safety-related equipment, but the loss of power created difficulties in carrying out routine maintenance and repair of some equipment. In an urgent letter to IAEA Director General Rafael Mariano Grossi, the SNRIU said: "The battle is going on in the town of Enerhodar and on the road" to the Zaporizhzhia nuclear plant. Grossi called for a halt to the use of force and called upon military forces to refrain from violence near the nuclear power plant.

4 March

Fighting between forces overnight resulted in a projectile hitting a training building within the site of the six-unit Zaporizhzhia plant. The SNRIU said a resulting fire at the site had not affected "essential" equipment and that plant personnel were taking mitigatory actions. There was no reported change in radiation levels at the plant and the six reactors were not affected. Russian forces took control of the plant. Energoatom informed the IAEA that the plant had been allowed to change work shifts.

5 March

The SNRIU reported to the IAEA that staff at the Chernobyl nuclear plant has been onsite since 23 February without being able to rotate shifts for either technical personnel or guards.

6 March

The SNRIU reported that staff at the Zaporizhzhia plant were now under orders from the commander of the Russian forces that took control of the site on 4 March. The regulator also said that Russian forces had switched off some mobile networks and the internet, preventing the communication of information from the site via normal channels.

9 March

The Chernobyl nuclear plant was disconnected from the electricity grid. The IAEA stated that it did not see a critical impact on safety as a result.

13 March

On 13 March Energoatom reported that transmission system operator Ukrenergo had at 18.38 succeeded in repairing a power line needed to restore external electricity supplies to Chernobyl.

14 March

At 16.45 local time the Chernobyl nuclear plant was reconnected to the national grid.

16 March

Ukraine and Moldova's electricity networks were connected to the continental European grid on a trial basis.

20 March

The SNRIU confirmed that around half of the staff at Chernobyl nuclear plant had been able to rotate and return to their homes for the first time since 23 February.

29 March

IAEA Director General Rafael Mariano Grossi entered Ukraine to provide technical assistance to ensure the safety and security of the country's nuclear facilities.

31 March

Energoatom said that Russian forces had left the Chernobyl site and nearby town of Slavutych.

7 April

In response to reported atrocities committed by the Russian armed forces, the European Parliament voted in favour (513 for; 22 against; 19 abstentions) of a resolution calling for a number of punitive measures against Russia, including “an immediate full embargo on Russian imports of oil, coal, nuclear fuel and gas”.

26 April

IAEA Director General Rafael Mariano Grossi arrived in Ukraine with experts from the agency to "conduct nuclear safety, security and radiological assessments, deliver vital equipment and repair the

Agency's remote safeguards monitoring systems" at the Chernobyl site.

22 July

Ukrainian military launched drone strikes on Russian forces stationed at the Zaporizhzhia nuclear plant.

5 August

Shelling reportedly damaged the Zaporizhzhia plant's external power supply system, triggering the emergency protection system of one of the three reactors (unit 4) operating at the time.

6 August

Shelling reportedly injured a Ukrainian security guard and damaged "walls, a roof and windows" in the area of the Zaporizhzhia plant's spent fuel storage facility.

25 August

Damage to an offsite power supply line forced operators at Zaporizhzhia to disconnect the plant's two operating units from the country's electrical grid. Offsite power supply was restored later the same day.

1 September

A team of inspectors from the IAEA arrived at the Zaporizhzhia plant. The IAEA assessed the physical damage to facilities, determined the functionality of the main and back-up safety and security systems, evaluated the working conditions of the control room staff, and undertook safeguards activities. IAEA Director General Rafael Mariano Grossi said that it was "obvious that the plant and the physical integrity of the plant has been violated several times." The IAEA planned to keep a permanent presence of two staff at the plant.

21 September

At 1:13 am, renewed shelling at the Zaporizhzhia plant damaged cables providing electricity to one of its six reactors. Unit 6 had to rely on two of the three emergency diesel generators for 40 minutes to ensure essential safety functions.

1 October

Ihor Murashov, Director General of the Zaporizhzhia plant, was detained by Russian forces. On 4 October the IAEA said it had been

informed that Murashov would not be returning to his role at the plant.

5 October

Russian President Vladimir Putin issued a decree to transfer the Zaporizhzhia plant to Russian control. Ukraine described the decree as "null and void".

6 October

Renewed shelling damaged a 150 kV power line providing electricity via the switchyard of a nearby thermal power station to Zaporizhzhia unit 6. Five emergency diesel generators started supplying back-up power to the unit. The generators operated for about an hour and a half, before an alternative source of power from four of the other reactors was connected.

8 October

The Zaporizhzhia plant lost access to external power following shelling. The plant's emergency diesel generators provided cooling and other essential nuclear safety and security processes. External power was restored on 9 October.

12 October

The Zaporizhzhia plant lost external power for a few hours following damage to a substation located far from the plant itself.

3 November

The South Ukraine plant lost connection to one of its three 750 kV power lines used to supply electricity to the grid.

15 November

The Khmelnitski plant lost connection to the grid and the Rivne plant lost connection to one of its 750 kV power lines following Russian missile strikes on Ukraine's energy infrastructure. Neighbouring Moldova experienced blackouts as a result of the strikes.

19&20 November

Shelling at the Zaporizhzhia plant damaged buildings, systems and some equipment.

23 November

All units at Rivne, South Ukraine and Khmelnitski were automatically disconnected from Ukraine's grid following a decrease in frequency resulting from Russian missile strikes on Ukraine's energy infrastructure. Neighbouring Moldova experienced blackouts as a result of the strikes.

22 December

Rosatom Director General Alexey Likhachev and IAEA Director General Rafael Mariano Grossi held talks over the creation of a nuclear and physical safety protection zone at Zaporizhzhia, with both parties outlining the aim to agree and implement the measure soon.

29 December

Shelling damaged the 330 kV Ferosplavna 1 power line connected to the Zaporizhzhia plant. At the time, this was the last functioning back-up power line.

2023

17 January

Teams of nuclear safety and security experts from the IAEA were stationed at Ukraine's nuclear power plants and the Chernobyl site as part of the IAEA's efforts to reduce the risk of a severe nuclear accident during the ongoing conflict. The IAEA expected to have around 11 or 12 staff in the country.

10 February

Renewed shelling of energy infrastructure resulted in precautionary reduced power output at the Rivne and South Ukraine nuclear power plants.

6 May

It was reported that residents from Enerhodar, a nearby town to Zaporizhzhia, had been evacuated.

6 June

The Nova Kakhovka hydroelectric dam was severely damaged during the early hours of 6 June, resulting in the draining of the Kakhovka reservoir. The reservoir behind the dam provided water for much of the south of Ukraine, and was the ultimate heat sink for the Zaporizhzhia nuclear power plant.

The Zaporizhzhia plant has a large cooling pond with sufficient water for several months and can draw water from the discharge channel of the nearby thermal power plant.

15 September

The IAEA announced that almost all the cooling needs for Zaporizhzhia's six offline reactors would be covered by nine new groundwater wells that supply the sprinkler ponds.

22 September

The IAEA announced that a total of ten groundwater wells had been drilled, and a further one planned, to provide a long-term solution to the provision of cooling water to the reactors.

13 October

It was announced that unit 5 was being brought into hot shutdown in order to provide hot water and district heating over the winter. Unit 4 was already in hot shutdown to generate steam for processing liquid radioactive waste and to heat water for nearby Enerhodar; the other reactors were in cold shutdown. The SNRIU had issued regulatory orders in June to bring all six units into cold shutdown.

2024

7 April

The IAEA announced that there had been three drone strikes at the Zaporizhzhia nuclear power plant resulting in minor superficial scorching to the top of the reactor dome roof of Unit 6 and scoring of a concrete slab supporting the primary make-up water storage tanks. There were reportedly no indications of damage to critical nuclear safety or security systems at the site.

15 April

The IAEA announced that all six units at Zaporizhzhya were in cold shutdown following the end of the winter heating season in the nearby town of Enerhodar.

12 August

The IAEA reported that its experts had witnessed "thick dark smoke" coming from one of the Zaporizhzhya plant's two cooling

towers. The IAEA confirmed that there was no impact on nuclear safety.

22 August

The IAEA reported that it had been informed by Russia that the remains of a drone were found within the territory of the Kursk nuclear power plant.

26 August

Widespread strikes targeting Ukraine’s energy infrastructure caused major fluctuations in power supply and led to the temporary shutdown or disconnection of reactor units at the Rivne and South Ukraine NPPs.

28 November

The IAEA reported that Khmelnitsky, Rivne and South Ukraine had to lower their power levels for the second time in two weeks in response to attacks on Ukraine's electricity grid.

13 December

Several units of the Khmelnitsky, Rivne and South Ukraine plants lowered power levels during widespread military activities in Ukraine.

16-23 December

IAEA experts visited electrical substations deemed critical to the safe operation of Ukraine's nuclear power plants. This followed two previous substation assessments by the Agency.

2025

14 February

A drone was reported to have struck the roof of the New Safe Confinement (NSC) structure at Chernobyl. The IAEA said that both the outer and inner cladding of the NSC's arch had been breached, but that radiation levels were stable.

30 September

Seven days after the Zaporizhzhia nuclear plant was disconnected from its last offsite power line on 23 September, Ukrainian President Volodymyr Zelensky warned that the situation at the plant had become critical after continued shelling prevented repairs

to be carried out. This was the longest such outage since the war began.

23 October

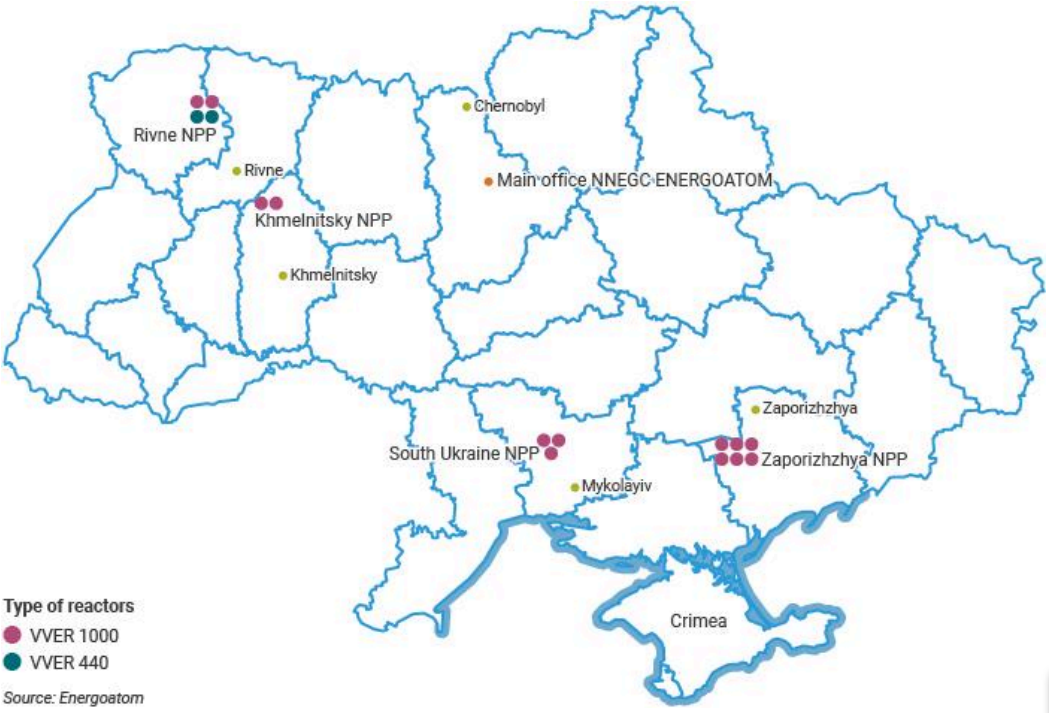
Following repairs carried out during a local ceasefire, the Zaporizhzhia nuclear plant was reconnected to the national electrical grid after a month-long outage. IAEA Director General Rafael Mariano Grossi said: "After exactly one month without any offsite power, the Zaporizhzhia nuclear power plant is once again receiving the external electricity it needs to cool its six reactors and spent fuel."

Ukraine nuclear power plants overview

Ukraine has 15 operable nuclear reactors at four plants that generate about half of its electricity. All reactors are Russian VVER types, two being upgraded 440 MWe V-312 models and the rest the larger 1000 MWe units – two early models and the rest V-320s.

Since the conflict started the number of units operated has changed over time, with reactors put online and taken offline depending on the situation around the plants and the stability of external power supplies.

For more information on the country's nuclear energy industry, see information page on Nuclear Power in Ukraine.



Zaporizhzhia nuclear power plant

2022

In the early hours of 4 March 2022, the plant became the first operating civil nuclear power plant to come under armed attack. A projectile hit a training building located approximately 300 metres from unit 1 of the plant (see image below). SNRIU stated that the resulting fire was extinguished at 06:20 local time on 4 March. The IAEA said that the fire at the site had not affected "essential equipment". Just one of the six reactors (unit 4) at the plant was producing electricity at the time of the attack. The statuses of unit 1 (maintenance outage) and units 5&6 (held in reserve, operating in low power mode) were initially unchanged as a result of the attack, whilst units 2&3 underwent a controlled shutdown. However, on 9 March Energoatom reported that unit 6 had been put in "emergency repair" on 7 March as a result of damage sustained to the block transformer during fighting in the early hours of 4 March.

Image 1: Fire at the training centre within the site of the Zaporizhzhia nuclear plant after fighting in the early hours of 4 March (Photo: Energoatom via AP)

On 5 March the SNRIU confirmed that the technical safety systems were intact and radiation levels remained normal. It reported that one telephone communication line had been lost but another was still functioning, as well as mobile phone communication. The IAEA was informed that the training centre hit by a projectile in the early hours of 4 March had suffered significant damage, and that there had also been damage to the site's laboratory building and to an administrative structure. A visual inspection of the dry storage facility did not detect any damage.

On 6 March the SNRIU said that staff at the Zaporizhzhia plant were now rotating in three shifts, but were under orders from the commander of the Russian forces that took control of the site on 4 March. According to the regulator, any action of plant management, including measures related to the technical operation of the plant, requires prior approval from the commander. The regulator also said that Russian forces had switched off some mobile networks and the internet, preventing the communication of information from the site via normal channels.

On 9 March the SNRIU informed the IAEA that two of the plant's five high-voltage offsite power lines had been damaged and so were inoperable. Energoatom said that the plant's offsite power needs could be provided with just one power line. The IAEA said that remote data transmission from safeguards monitoring systems installed at the facility had been lost.

On 11 March the SNRIU said that staff at the plant were rotating according to their usual schedule and without interference in their day-to-day activities. The IAEA reported that remote data transmission from safeguards systems at the plant was back online, having been lost on 9 March.

On 14 March Energoatom said that Russian forces had carried out munition explosions at the site. The SNRIU had previously informed the IAEA about work to detect and dispose of any unexploded munitions following the events on 4 March when Russian forces took control of the site.

On 16 March the SNRIU said that the Zaporizhzhia plant had lost connection to a third power line, leaving two available. The regulator has said that a single power line is sufficient to provide power for all safety systems.

On 17 March a break in an onsite power line prompted the operators of the plant to temporarily reduce the output of the two operating reactors slightly from 600 MWe to 500 MWe each. The onsite line was fixed later on the same day and power levels were increased.

On 19 March the SNRIU said that one of the three disconnected offsite power lines had been restored. The repair of the line means that the site now has three offsite power lines available, including one held in reserve.

On 27 March the SNRIU said that repair work to the transformer of unit 6, which was damaged during the events of 4 March, had been completed.

In its report released on 28 April, the IAEA said that "about ten" Rosatom staff members were present at the Zaporizhzhia plant and that it "considers the presence of Rosatom senior technical staff could lead to interference with the normal lines of operational command or authority, and potential frictions when it comes to decision-making."

On 30 May the automatic transfer of safeguards data to the IAEA from systems installed at the Zaporizhzhia plant stopped. The IAEA

said on 12 June that transmission of the data had restarted. It was lost again on 25 June before being restored on 4 July.

On 22 July Ukraine's military intelligence service released footage of a drone strike on Russian forces at the Zaporizhzhia plant.

On 3 August Grossi said the IAEA had limited communication with the workers at the plant and reiterated the need for the IAEA to travel to Ukraine to inspect the plant.

On 5 August shelling reportedly damaged one of the plant's external power supply lines. Ukraine informed the IAEA on 6 August that there had been no damage to the reactors themselves, no radiological release and no reports of injuries, and that two external power lines remained operational. Damage was reported to an auxiliary system, specifically a nitrogen-oxygen station, and an unspecified auxiliary building.

On 6 August shelling reportedly damaged "walls, a roof and windows" in the area of the plant's spent fuel storage facility and injured a Ukrainian security guard.

On 7 August Energoatom said that a worker at the plant had been wounded during the shelling on 5 August.

Further shelling was reported to the IAEA by Ukraine on 11 August. Ukraine said the shelling had caused damage to radiation monitoring equipment at the plant's fire station.

On 23 August Ukraine reported to the IAEA that further shelling on 20 and 21 August had damaged laboratory and chemical facilities in the vicinity of the nuclear plant, and damaged transformers of the non-nuclear thermal plant that is situated about 3.5km east of the nuclear reactors. The damage to the non-nuclear plant caused a disconnection of the power line linking it to the nuclear plant that lasted for several hours.

On 25 August Ukraine informed the IAEA that the plant lost connection to its last remaining operational external power line. As a result, the two reactors (units 5&6) that were operating at the plant were disconnected from Ukraine's electricity grid. The power line connection was restored later on 25 August, and reactors 5&6 were reconnected to the grid on 26 August.

The power needs of the six-unit plant are normally served by four 750 kV external power lines connected to Ukraine's grid, with three 330 kV lines, connected to a nearby thermal power plant, providing back-up. For more information on the plant's external and back-up power supplies, see below.

On 1 September a team of inspectors from the IAEA arrived at the Zaporizhzhia plant following months of negotiations. The IAEA visited the site to assess the physical damage to facilities at the plant, determine the functionality of the main and back-up safety and security systems, evaluate the working conditions of the control room staff, and undertake safeguards activities. IAEA Director General Rafael Mariano Grossi said that it was "obvious that the plant and the physical integrity of the plant has been violated several times." The IAEA planned to keep a permanent presence of two staff at the plant.

On 3 September the plant once again lost connection to its last operational 750 kV power line following further shelling in the area. The IAEA said that one of the three 330 kV reserve lines was delivering the electricity generated by the Zaporizhzhia plant to the grid – and that the same line can provide back-up power to the plant if needed. Ukraine said the 750 kV line would be repaired but that it would take several days to do so.

On 5 September the 330 kV reserve line being used was deliberately disconnected due to having to extinguish a fire. The line itself was not damaged, and electricity generated at unit 6 in low power mode was used to provide the plant with its safety and security needs. Ukraine said that the 330 kV line would be reconnected once the fire has been extinguished.

On 6 September the IAEA published its report[3] on the situation at Zaporizhzhia following its visit to the site. The IAEA reported damage to the following caused by the events since March:

- One turbine lubrication oil tank.
- The roofs of various buildings such as the building for the spent fuel transporter vehicle.
- The special building that houses, among other items, the fresh nuclear fuel and the solid radioactive waste storage facility.
- The new training building.
- The building where the central alarm station of the physical protection system is located.
- The container where the radiation monitoring system is located, in the vicinity of the dry spent fuel storage facility.

The IAEA noted: "The presence of Russian military personnel, vehicles and equipment at various places [at the plant] including several military trucks on the ground floor of the unit 1 and unit 2 turbine halls and military vehicles stationed under the overpass connecting the reactor units."

Following its visit the IAEA called for the "immediate establishment of a nuclear safety and security protection zone."

Image 2: IAEA director general Grossi inspects shelling damage during his visit to the plant (Source: IAEA)

Further shelling on 7 September damaged one of the three 330 kV reserve lines, and then the thermal power plant's switchyard on 8 September, causing a complete power blackout in the local city of Enerhodar.

On 10 September the damaged 330 kV reserve line was restored, and connection re-established, providing the nuclear plant with the electricity it needs for safety and security functions directly from the grid. Following restoration of this line, unit 6, which had been operating in low power mode to provide the plant with its electricity needs, was shut down.

On 21 September shelling damaged cables providing electricity to unit 6 at the plant. Two of the unit's three emergency diesel generators provided power to the unit for about 40 minutes whilst access to external electricity supplies were re-established.

In the week commencing 26 September a number of landmines exploded near the plant, likely caused by the movement of wildlife in the local area. On 27 September the IAEA said that explosions had broken windows in the turbine hall of unit 2. On 30 September the Agency said that an explosion had damaged a 6 kV cable just

outside the plant's perimeter fence. Due to a short-circuit of the damaged cable, the explosion indirectly damaged a voltage transformer at unit 6, resulting in smoke at the site.

On 1 October Ihor Murashov, Director General of the Zaporizhzhia plant, was detained by Russian forces. On 4 October the IAEA said it had been informed that Murashov would not be returning to his role at the plant. The IAEA said it was not yet clear who would replace him.

On 5 October Russian President Vladimir Putin issued a decree to transfer the Zaporizhzhia plant to Russian control and oversight. The decree said that a new company, the Operating Organisation of Zaporizhzhia NPP, has been created with the "status of a nuclear energy operating organisation to operate and/or discontinue the operation of nuclear power facilities at the Zaporozhye Nuclear Power Plant." Rosatom in a statement on its website said that the new company would be headed by Oleg Romanenko, who was previously chief engineer of the Balakovo nuclear power plant. Ukraine's Ministry of Foreign Affairs described the statement as an "illegal attempt" to transfer operational control of the plant and said it was the latest in "a long line of crimes and violations of international law committed by Russia." It added that it considered the decree to be "null and void".

Also on 5 October the IAEA said that its experts had learned that unit 5 was planned to be restarted at reduced power to produce steam and heat for the plant.

On 8 October the plant lost access to external power after shelling damaged the last operating 750 kV power line. Emergency diesel generators provided cooling and other essential nuclear safety and security processes. In response, IAEA Director General Rafael Mariano Grossi described the renewed shelling as "tremendously irresponsible" and emphasized the need for a "nuclear safety and security protection zone around the plant." External power was restored on 9 October.

On 12 October external power was lost again for a few hours following damage to a substation located far from the plant itself.

On 14 October one of three backup power lines connecting the Zaporizhzhia plant to a thermal power plant 3.5 km east of the reactors was restored. This line can be used to provide external power if required.

On 18 October connection to the plant's last operating 750 kV power line was restored.

On 27 October the IAEA reported that the Zaporizhzhia plant had over the past ten days been receiving power from the national grid without interruption. It also said that a second backup power line connecting the plant to a nearby thermal plant had been restored in recent days.

Also on 27 October the IAEA highlighted changes to the operational command at the facility since Russia's decree on 5 October. It said that the number of Russian technical staff at the site had increased and that they were now involved in "taking significant operational decisions". For example, Ukrainian operating staff had planned to restart unit 5, but it currently remains in a hot shutdown mode as Russian officials have not agreed to start it up again.

On 3 November the plant lost connection to its 750 kV power line. Hours later it also lost connection to its 330kV reserve line. Ukrainian operating staff said that damage to the lines had been sustained about 50-60 kilometres from the plant itself. External power was restored on 5 November. On 19 and 20 November repeated shelling at the Zaporizhzhia plant resulted in damage to "buildings, systems and equipment" according to IAEA. Site management reported damage to a radioactive waste and storage building, cooling pond sprinkler systems, an electrical cable to one of the diesel generators, condensate storage tanks, and to a bridge between a reactor and its auxiliary buildings. The plant's external power supplies were not impacted.

On 22 December Rosatom Director General Alexey Likhachev and IAEA Director General Rafael Mariano Grossi held talks over the creation of a nuclear and physical safety protection zone at Zaporizhzhia, with both parties outlining the aim to agree and implement the measure soon.

On 29 December shelling on the other side of the Dnipro river damaged the 330 kV Ferosplavna 1 power line – the last functioning back-up power line connected to the Zaporizhzhia plant. IAEA Director General Rafael Mariano Grossi said it had to be restored as soon as possible and that he was continuing his consultations aimed at implementing a nuclear safety and security protection zone around the plant as soon as possible.

2023

On 17 January after repeated cuts, the Ferosplavna 1 back-up line was reconnected to the Zaporizhzhia plant. The plant continued to

receive offsite power for nuclear safety and security functions from its last remaining operational 750 kV power line.

On 3 February IAEA Director General Rafael Mariano Grossi raised concerns over a recent reduction in the water level of the Kakhovka reservoir, which forms part of the Dnipro river and provides cooling water for the Zaporizhzhia plant. The large cooling pond next to the Zaporizhzhia plant is designed to stay above the level of the Kakhovka reservoir, meaning that the decreased water level does not pose an immediate threat to plant operations.

On 10 February Grossi met with senior Russian government officials in Moscow as part of ongoing efforts to establish a nuclear and physical safety protection zone at Zaporizhzhia. It was also reported that water levels in the Kakhovka reservoir had fallen to their lowest in 30 years, with Ukraine's hydroelectric company, Ukrhydroenergo, claiming that Russian military forces were deliberately discharging water from the reservoir.

On 1 March IAEA Director General Rafael Mariano Grossi expressed concerns over a more than three-week delay in the rotation of IAEA experts at Zaporizhzhia. The continued conflict around Zaporizhzhia resulted in the disconnection of the plant's last operational backup power line.

On 2 March Grossi announced that rotation of IAEA staff had been completed.

On 9 March the 750 kV line was disconnected, forcing the plant to rely on backup diesel generators for 11 hours.

On 23 March the IAEA announced that the 330 kV backup power line damaged on 1 March remained disconnected and under repair, forcing the plant to rely on the only remaining 750 kV external power line.

On 29 March Grossi arrived at the Zaporizhzhia plant for the second time during the conflict. Grossi stated his intention to review the situation at the plant and talk with the operators, whilst also acting as a guarantor for the rotation of IAEA staff working at the facility. Grossi announced plans to make some adjustments to the 'nuclear safety and security protection zone' proposal in order to establish some form of protection.

On 20 April Rosenergoatom announced that it woudl change Zaporizhzhia's fuel from US-made to Russian fuel, following a report that the US Department of Energy had sent a letter saying that the occupied plant in Ukraine "contains US-origin nuclear

technical data that is export-controlled by the United States Government".

On 25 April plant management informed the IAEA that the scope of maintenance performed during outages on all units in 2022 was reduced compared with the planned scope, due to reduced maintenance staff, absence of external contractors who perform a significant part of the work, and a lack of spare parts needed for the maintenance, including critical components. The IAEA said that the plant had only about one-quarter of its regular maintenance staff available.

On 6 May IAEA experts received information that residents from nearby town of Enerhodar had been evacuated. The Zaporizhzhia plant director confirmed that operating staff would not be evacuated. IAEA Director General Rafael Mariano Grossi said: “I’m extremely concerned about the very real nuclear safety and security risks facing the plant. We must act now to prevent the threat of a severe nuclear accident and its associated consequences for the population and the environment.”

On 30 May Grossi provided a statement to the United Nations Security Council, in which he outlined five "concrete principles" to ensure nuclear safety and security at the Zaporizhzhia plant. These principles follow consultation with the leadership of Ukraine and Russia. They are:

1. There should be no attack of any kind from or against the plant, in particular targeting the reactors, spent fuel storage, other critical infrastructure, or personnel.
2. The plant should not be used as storage or a base for heavy weapons (i.e. multiple rocket launchers, artillery systems and munitions, and tanks) or military personnel that could be used for an attack from the plant.
3. Offsite power to the plant should not be put at risk. To that effect, all efforts should be made to ensure that offsite power remains available and secure at all times.
4. All structures, systems and components essential to the safe and secure operation of the plant should be protected from attacks or acts of sabotage.
5. No action should be taken that undermines these principles.

On 6 June the Nova Kakhovka hydroelectric dam was severely damaged, resulting in the draining of the Kakhovka reservoir. The reservoir behind the dam provided water for much of the south of Ukraine, and was the ultimate heat sink for the Zaporizhzhia plant.

The plant's large cooling pond combined with the discharge channel of the nearby thermal power plant can provide for its cooling needs for several months. Additonal cooling water can be

pumped from an underground water system and on-site wells if required.

On 8 June Ukraine's nuclear regulator ordered for unit 5 to be placed into 'cold shutdown'. Unit 5 had been in hot shutdown to provide steam to support processes that contribute to safety at the site. The IAEA has said it was encouraging the plant to investigate the possibility of installing an external boiler to generate the required steam so that all units could be moved to cold shutdown.

On 1 July the plant restored connection to its only available 330kV backup power line which was lost on 1 March.

On 12 July IAEA described the water levels in the two main bodies of water available to the plant – its cooling pond and the discharge channel of the nearby thermal plant – as "relatively stable." The water level was decreasing by 1-2 centimetres per day due to usage and evaporation.

Earlier on 22 June the Ukrainian president, Volodymyr Zelenskyy, said that Russia was preparing to commit a "terrorist attack" at the Zaporizhzhia plant. On 12 July the IAEA, which has permanent representation at the plant, said it continued inspections "without seeing any mines or explosives." The Agency stated that it was still waiting to gain access to the rooftops of the turbine buildings of units 3&4 after reports of objects being placed there.

Also on 12 July the IAEA said that the plant was preparing to move unit 4 from cold shutdown to hot shutdown, after which unit 5, then in hot shutdown, would be placed in cold shutdown. This was to allow maintenance activities to be performed on unit 5.

On 20 July the IAEA said its experts had carried out additional inspections and walkdowns at the Zaporizhzhia plant without observing any heavy military equipment, explosives or mines. The Agency said it was still awaiting access to the rooftops of the reactor buildings.

On 22 August the IAEA said that the Zaporizhzhia plant had started pumping water from a new groundwater well and planned to build more. The new well was providing about 20 m3 of water per hour. Around 10-12 additional wells were planned around the perimeter of the sprinkler ponds. The Agency confirmed that the site still had cooling water supply for many months.

On 15 September the IAEA announced that there was a total of nine new groundwater wells, which together could provide almost

all of the cooling needs for the six reactors and used fuel, pumping around 200 m3/h into the sprinkler ponds.

On 22 September the IAEA announced that a total of ten groundwater wells had been drilled to provide a long-term solution to the provision of cooling water to the reactors. Drilling of one additional well was planned, which would bring the total to 11 and provide the necessary 250 m3/h of water to maintain the cooling of the reactors and spent fuel pools.

On 30 September following the repair of a water leak, unit 4 was returned to hot shutdown. Unit 6 – which was temporarily producing the necessary steam – was put back into cold shutdown.

On 3 October a new team of IAEA experts arrived at the Zaporizhzhia power plant to replace staff who had been there for several weeks.

On 13 October the IAEA was informed that unit 5 was being brought into hot shutdown in order to provide hot water and district heating over the winter. This was the second reactor to be in hot shutdown, along with unit 4, whist the other reactors were in cold shutdown.

2024

On 7 April the IAEA announced that there had been three drone strikes at the Zaporizhzhia nuclear power plant. While there were no indications of damage to critical nuclear safety or security systems at the site, IAEA experts identified that one of the impacts was at the building of unit 6, where surveillance and communication equipment was seemingly targeted. The experts confirmed that there was minor superficial scorching to the top of the reactor dome roof of Unit 6 and scoring of a concrete slab supporting the primary make-up water storage tanks. This was the first time the plant had been directly targeted in military action since November 2022.

On 11 August IAEA experts reported hearing multiple loud explosions near the Zaporizhzhia nuclear power plant. Following the explosions the Agency's experts witnessed large plumes of dark smoke coming from one of the plant's two cooling towers, located to the north-west, outside the plant's perimiter. The IAEA said that there was no direct safety risk. Walkdowns in the days that followed showed extensive damage to internal components at the water nozzle distrubtion level of the cooling tower, about 10

metres above ground level. The IAEA concluded that the fire was unlikely to have started at the base of the cooling tower, as had previously been suggested.

On 17 August a drone with an explosive payload detonated just outside the plant's perimeter. There were no casualties and no impact on any plant equipment, but the road between the two main gates of the plant was damaged.

In the week commencing 14 October the IAEA team at the Zaporizhzhia plant was informed that two of the six mobile diesel generators introduced following the Fukushima stress tests had been moved to other locations. One was now being used at the recently damaged Zaria substation and one at the pumping water supply station in the city of Enerhodar.

On 29 October the IAEA team visited unit 1 after being informed that one of the impulse lines, part of the reactor coolant pump support systems, was leaking and required repair. The required work was completed by 31 October. The unit returned to cold shutdown mode on 2 November.

On 10 December a drone hit an IAEA vehicle involved in the transport of Agency staff to and from Zaporizhzhia plant. The vehicle was severely damaged but the two occupants were unharmed.

In week commencing 9 December Agency staff observed three new mobile diesel generators located adjacent ot turnine buildings of three of the reactor units. The ZNPP staff informed the IAEA team that these were in accordance with the regulations of the Russian Federation as as part of post-Fukushima Daiichi accident measures and in addition to those previously implemented by Ukraine. In late 2022, the ZNPP received seven other mobile diesel generators, which are now disconnected and not in use.

2025

On 5 January a drone attack on the Zaporizhzhia plant's training center was reported. The training center is located just outside the perimeter of the plant.

On 23 January the IAEA team at the plant was informed that three new mobile diesel generators were to be delivered to the site, similar to those received late in 2024. The team were informed that

four diesel steam generators were put into operation for ten days to provide the steam needed to process liquid radioactive waste.

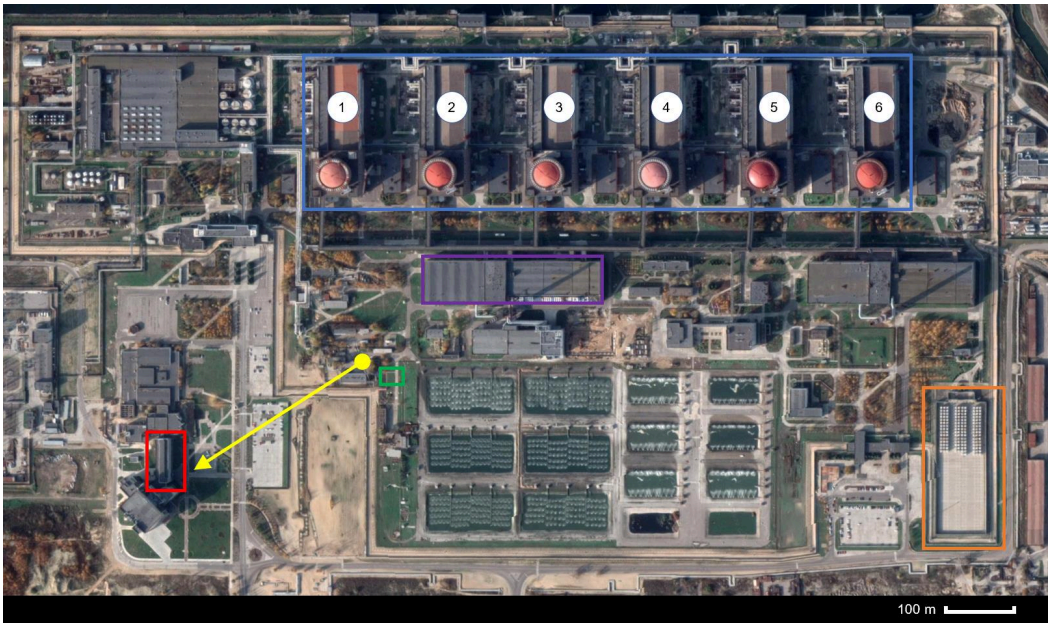


Image 3: location of reactors, spent fuel storage facility, and key events on 4 March and 22 July
Blue rectangle and numbered – the six units of the Zaporizhzhia nuclear power plant
Red rectangle – location of the training centre hit by a projectile on 4 March
Yellow arrow – approximate location and direction of Image 1
Orange rectangle – location of the dry storage facility for used fuel
Green rectangle – location of Ukrainian drone strike 22 July
Purple rectangle – area of shell impacts recorded by the IAEA on its visit to site
Sources: Google, IAEA, Defence Intelligence of Ukraine

The reactors at the six-unit Zaporizhzhia nuclear power plant are Russian-designed VVER-1000 (V-320) units commissioned between 1984 and 1995. The V-320 has a full-pressure single containment, which has the shape of a hollow cylinder with a spherical dome and a flat bottom. The reinforced concrete wall of the containment is 1.2m thick in its cylinder-shaped section and 1.1m thick in its dome section. There is an 8mm steel liner on the internal side of the containment.

Spent fuel assemblies removed from each reactor at Zaporizhzhia are initially stored for several years in racks in spent fuel pools inside the reactor containment. Once sufficiently cooled, fuel assemblies are transferred to a dry spent fuel storage facility situated at the site. At Zaporizhzhia, the dry storage facility is protected along its perimeter by a concrete wall 30cm thick and 6m

high. The spent fuel stored at the dry storage facility is in concrete containers.

External and back-up power supply

Zaporizhzhia has four main high voltage (750 kV) offsite power lines. Three 330 kV back-up lines, connected to a thermal power plant 3.5 km east of the reactors, remain on standby during normal operation, but can be used to deliver the electricity generated at Zaporizhzhia to the grid, and to provide back-up power for the plant. In the event of complete loss of offsite power, each reactor has three back-up diesel generators. One generator is sufficient to maintain a reactor and its spent fuel in a safe state for ten days. A further two 'bunkerized' diesel generators are present onsite. Each unit at the plant is equipped with one mobile diesel generator, which could be used to provide power to any of the other units if needed.[1]

Nuclear power plants require back-up electricity supplies to provide cooling for the removal of decay heat produced by shutdown reactors, and to maintain services (e.g. systems control, lighting, communication, ventilation) to the site.

Geneva Conventions

The Additional Protocol of 1979 to the Geneva Conventions contains in Article 56 a provision stating that nuclear power plants “shall not be made the object of attack, even where these objects are military objectives, if such an attack may cause the release of dangerous forces and consequent severe losses among the civilian population."

Emergency preparedness

In September 2017, the State Emergency Service of Ukraine in cooperation with the Defence Threat Reduction Agency of the United States Department of Defense and the Federal Emergency Management Agency of the USA conducted an emergency response exercise related to a radiation accident at the Zaporizhzhia nuclear power plant. One of the main tasks of this training was to improve plans and procedures to protect the public against natural, man-made, social and military threats. Further exercises were carried out with support of the US Department of

Defense in 2018 to check the effectiveness of the facility's plans in the event of sabotage and anti-terrorism activity.[2]

In December 2022 IAEA Director General Rafael Mariano Grossi and Ukrainian prime minister Denys Shmyhal agreed on a continued IAEA presence at all of the country's nuclear power plants.

South Ukraine nuclear power plant

2022

On 16 April the SNRIU informed the IAEA that onsite video surveillance had recorded a missile flying directly over the South Ukraine nuclear plant. The IAEA said it was "looking into this matter, which, if confirmed, would be extremely serious."

On 5 June Energoatom said it had recorded a Russian missile flying "critically low" over the South Ukraine nuclear plant.

On 20 September Energoatom reported that a Russian missile landed 300 metres away from the reactors at the South Ukraine nuclear plant (also known as Pivdennoukrainsk Nuclear Power Plant, PNPP), causing damage to the plant. Energoatom stated: "All three power units of the PNPP are operating normally [with] no casualties among the station staff."

On 3 November the South Ukraine plant lost connection to one of its three 750 kV power lines used to supply electricity to the grid. As a result the plant reduced the power of one of its three operating reactors by 50%.

On 23 November Russian missile attacks on Ukrainian energy infrastructure caused widespread blackouts across the country. A resulting drop in grid frequency caused reactors at the South Ukraine plant to automatically disconnect from the grid. The reactors were reconnected to the grid and were operating by 25 November.

2024

On 26 August, widespread strikes targeting Ukraine’s energy infrastructure caused major fluctuations in the power supply and

led to the temporary shutdown or disconnection of reactor units at the Rivne and South Ukraine NPPs. South Ukraine nuclear plant lost connection to two of its offsite power lines.

On 18 September the IAEA team at South Ukraine nuclear plant was informed that 22 drones flew through the area around the plant, including one over the plant.

On 21 September South Ukraine 2 was transferred to a cold shutdown state to repair the electrical motor of one of its main cooling pumps, which was damaged during the attacks on 26 August. The unit restarted on 25 September.

Khmelnitski nuclear power plant

2022

On 15 November the Khmelnitski nuclear plant lost all access to Ukraine's electricity grid. The plant's four operating power lines were progressively lost over a two-and-a-half-hour period as a result of Russian missile strikes on Ukraine's energy infrastructure. As a result of the strikes the two reactors were shut down and diesel generators provided required cooling. Offsite power was reestablished nine hours later through two 330 kV backup power lines, allowing the diesel generators to be turned off.

On 23 November Russian missile attacks on Ukrainian energy infrastructure caused widespread blackouts across the country. A resulting drop in grid frequency caused reactors at the Khmelnitski plant to automatically disconnect from the grid. The reactors were reconnected to the grid and were operating by 25 November.

Rivne nuclear power plant

2022

On 15 November connection to one of two 750 kV power lines linking Rivne nuclear plant to Ukraine's electricity grid was lost. As a result the plant reduced its power output, and one of its four units was automatically disconnected from the grid.

On 23 November Russian missile attacks on Ukrainian energy infrastructure caused widespread blackouts across the country. A resulting drop in grid frequency caused reactors at the Rivne plant to automatically disconnect from the grid. The reactors were reconnected to the grid and were operating by 25 November.

2024

On 26 August, widespread strikes targeting Ukraine's energy infrastructure caused major fluctuations in the power supply and led to the temporary shutdown or disconnection of reactor units at the Rivne and South Ukraine NPPs

Chernobyl

2022

On 24 February 2022 Russian forces took control of all facilities of the Chernobyl nuclear plant. Control levels of gamma radiation dose rates in the Chernobyl exclusion zone were exceeded. The SNRIU said that the rise in radiation levels was likely due to "disturbance of the top layer of soil from movement of a large number of heavy military machinery through the exclusion zone and increase of air pollution." It added: "The condition of Chernobyl nuclear facilities and other facilities is unchanged." Radiation readings from the site were assessed by the IAEA to be low and in line with near background levels.

On 2 March the SNRIU said it maintained communications with the site and that the site's personnel were carrying out their duties under "supervision".

On 5 March the SNRIU said that staff at the Chernobyl plant had been onsite since 23 February. As of 7 March the SNRIU said that there had still not been a rotation of technical personnel or guards since the Russian forces took control of the site.

On 8 March the SNRIU said that the handling of nuclear material at Chernobyl had been put on hold. It added that it could only communicate with the plant via email. The IAEA said that remote data transmission from safeguards monitoring systems installed at Chernobyl had been lost.

On 9 March at 11.22 local time the Chernobyl plant lost connection to the grid. The SNRIU said that back-up diesel generators were running and had 48 hours of fuel. The IAEA stated that, based on the heat load of spent fuel in the ISF-1 storage pool, and the volume of cooling water it contained, there would be sufficient heat removal without electrical supply. It said that it saw no critical impact on safety as a result of the loss of power, but said that the loss of power would likely create additional stress for the about 210 staff who have not been able to rotate for the past two weeks.

Professor Geraldine Thomas, director of the Chernobyl Tissue Bank, said: "They [the used fuel bundles] will not be producing significant amounts of heat, making a release of radiation very unlikely. In the unlikely event of a release of any radiation, this would be only to the immediate local area, and therefore not pose any threat to western Europe – there would be no radioactive cloud."

On 10 March the SNRIU told the IAEA that it had lost all communications with the Chernobyl nuclear plant. Prior to the loss of communication, the regulator had been able to confirm that the structures and systems of the ISF-1 used fuel storage pool had sustained no damage. The IAEA reiterated on 10 March that the loss of offsite power will not have a critical impact on essential safety systems. If emergency power to the site is also lost, the regulator said that staff would still be able to monitor the water levels and temperature of the spent fuel pool, albeit under "worsening radiation safety conditions" due to a lack of ventilation at the facility.

On 11 March the SNRIU said that technicians had started work on the evening of 10 March to repair damaged power lines to restore external electricity supplies. Emergency diesel generators continued to provide power to the site and additional fuel had been delivered.

European stress tests in 2011 included power loss scenarios at Chernobyl. The tests concluded that in the event of a loss of offsite power *and* back-up power, the water temperature in spent fuel pools would “increase but not exceed 70°C”.

On 13 March Energoatom reported that transmission system operator Ukrenergo had at 18.38 succeeded in repairing a power line needed to restore external electricity supplies to Chernobyl. However, Ukrenergo reported in the morning of 14 March that the line had sustained further damage "by the occupying forces". Later on 14 March Ukrenergo said that external power had been restored at 13.10 local time, and at 16.45 the plant was reconnected to Ukraine's electricity grid.

On 23 March the SNRIU reported that firefighters were trying to extinguish wildfires near the Chernobyl site. Seasonal wildfires often occur within the Chernobyl exclusion zone. The regulator and the IAEA assessed the radiological risk as low based on years of experience of such fires and on the amount of residual radioactive contamination in the soil.

On 24 March the State Agency for the Management of the Exclusion Zone said that an environmental laboratory had been "looted by marauders". The IAEA said that the incident did not pose a significant radiological risk.

On 30 March reports emerged of Russian forces receiving high doses of radiation whilst being in the Chernobyl exclusion zone. The IAEA said it had been unable to confirm the validity of these reports and that it was seeking further information to provide an independent assessment.

On 31 March the SNRIU told the IAEA that the Russian forces that had been in control of the Chernobyl plant since 24 February had, in writing, transferred control of the plant to Ukrainian personnel. Ukraine's defence ministry confirmed on 1 April that Russian forces had fully withdrawn from the Chernobyl plant.

On 19 April SNRIU informed IAEA that direct communications between it and the Chernobyl plant had been restored. Contact had been lost more than one month earlier during the period when Russian forces controlled the site.

From 25 to 28 April IAEA Director General Grossi led an IAEA mission to the Chernobyl site to "conduct nuclear safety, security and radiological assessments, deliver vital equipment and repair the Agency's remote safeguards monitoring systems." As part of the mission, IAEA technicians upgraded the remote radiation monitoring systems installed at the site and deployed new satellite transmission channels. Remote data transmission had been disrupted since the Russian forces took control of the site in February, but was restored on 11 May. A second mission was sent by the IAEA to the Chernobyl site between 31 May and 3 June.

2023

On 18 January teams of nuclear safety and security experts from the IAEA were stationed at the Chernobyl site as part of the IAEA support and assistance mission.

The Chernobyl plant was the site of a major nuclear accident in 1986. It was the product of a flawed Soviet reactor design coupled with serious mistakes made by the plant operators. It was a direct consequence of Cold War isolation and the resulting lack of any safety culture. The accident destroyed the Chernobyl 4 reactor. Chernobyl unit 4 was enclosed in a large concrete shelter which was erected quickly (by October 1986) to allow continuing operation of the other reactors at the plant. This structure was neither strong nor durable and was replaced by the New Safe Confinement (NSC) structure in 2017. Units 2, 1 and 3 were shut down in 1991, 1996 and 2000 respectively. There are no operating reactors at the site. For more information about the accident at Chernobyl, see information page on Chernobyl Accident 1986.

There are two fuel storage facilities at the Chernobyl plant. A wet storage pool (ISF-1) and a new, dry-storage facility (ISF-2), commissioned in April 2021. Fuel is being progressively transferred to the new facility. At present, about 2000 of the about 20,000 used fuel assemblies have been transferred to ISF-2.

2025

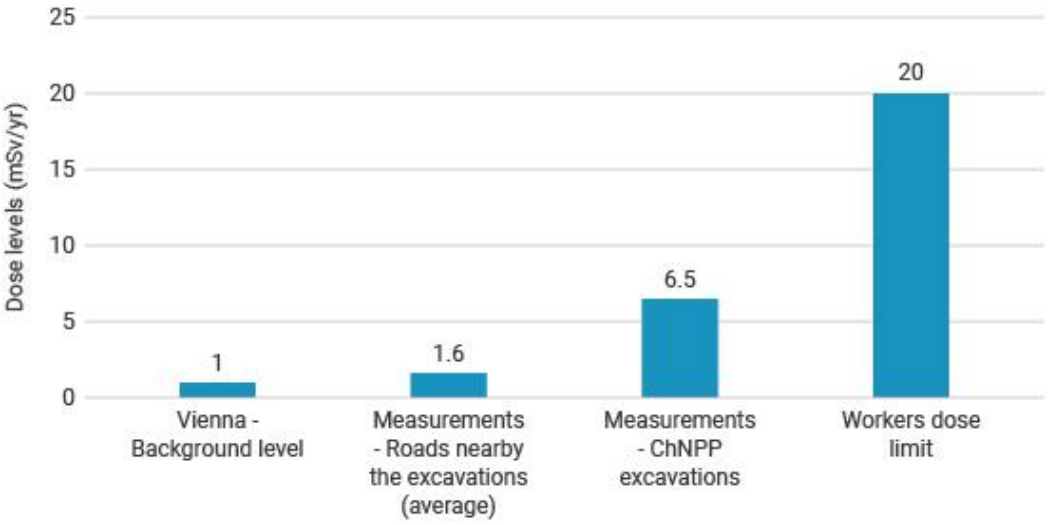
On 14 February a drone struck the NSC. The IAEA said that both the outer and inner cladding of the NSC's arch had been breached, but that radiation levels were stable. The Agency described the damage as "significant". Its onsite team observed a hole measuring approximately six metres in diameter, as well as damage to equipment and some electrical cables.

On 5 March the Agency said that firefighters had made progress extinguising the fire. Its team said that no smouldering fires had been detected over the past two days. On 13 March the Agency reported that firefighters had "gained full control" over the situation.

Radiation levels

As part of its mission from 25 to 28 April 2022, the IAEA assessed radiation levels in the Chernobyl exclusion zone, including in the excavations made by the Russian forces. The results ranged from an average of 1.6 mSv/yr on roads near to the excavations to 6.5 mSv/yr in the excavations themselves. Typical exposures to background radiation for people worldwide range from about 1.5 - 3.5 mSv/yr, but can be more than 50 mSv/yr, depending on geology, altitude and other factors. Workers at the Chernobyl site have an

authorized annual dose limit of 20 mSv/yr, three times higher than that recorded by the IAEA in the excavations.



Radiation dose levels recorded by the IAEA in the Chernobyl exclusion zone in April 2022, typical worldwide annual exposure from background radiation, and the authorized limit for workers at the Chernobyl nuclear plant (source: IAEA)

Staff/shift rotation

The SNRIU reported on 20 March 2022 that about half the staff at the Chernobyl site had been able to rotate and return to their homes for the first time since 23 February 2022. A day later the regulator confirmed that the remainder of the staff had been able to rotate and return home too. The new work shift included two supervisors instead of the usual one to ensure there was back-up available at the site.

A second staff rotation took place on 10 April 2022. The staff had to be transported by boat on the Pripyat River, which IAEA Director General Rafael Mariano Grossi said "underlined that the situation... remained far from normal."

On 21 April 2022 the SNRIU told IAEA that staff rotations were "taking place regularly and according to plan."

Non-power facilities

The SNRIU reported on 6 March 2022 that a subcritical assembly at the Kharviv Institute of Physics and Technology came under fire. A substation was destroyed, and air conditioning systems and heating lines were damaged.

The facility, known as Neutron Source, is used for research and to produce radioisotopes for medical and industrial applications. The facility is always subcritical* and contains very little radioactive material. As such, the IAEA concluded that the damage reported to it would not have had any radiological consequences.

The facility was put into a “deep subcritical state” on 24 February 2022.

The IAEA inspected the facility during the week commencing 7 November 2022. It reported that whilst the facility was heavily damaged by shelling, there was no indication of radiological release or diversion of declared nuclear material.

* A subcritical fission device does not release a sufficient number of neutrons to sustain an ongoing series of reactions.

Kursk nuclear power plant (Russia)

On 22 August 2024 the IAEA was informed that the remains of a drone had been found roughly 100 metres from the Kursk plant's spent fuel storage facility. Ukrainian troops advanced into the Russian region of Kursk in a surprise attack in early August. Claimed limits of the Ukrainian advance as of 22 August were approximately 30-40km from the Kursk nuclear power plant.

Fuel cycle implications

Russia has been an exporter of nuclear fuel services since 1973, and today occupies a significant role in the world market, particularly for enrichment services.

In response to Russia’s military offensive, western countries have sought to diversify their supply of nuclear fuel and fuel services.

Uranium trade

In May 2024 US President Joe Biden signed the Prohibiting Russian Uranium Imports Act. The legislation bans the import of Russian-produced unirradiated LEU as well as unirradiated LEU that has been swapped for banned uranium.

The bill came into effect on 11 August, and will last until the end of 2040. Waivers may be granted to allow the import of limited

amounts of LEU until 1 January 2028 if one of the following criteria is met:

- The import is necessary to maintain the viability of a US nuclear energy company that is critical to the US nuclear energy fuel supply chain.
- The import is intended to support an existing arrangement to provide fuel for a nuclear power plant in another country and thus minimize the likelihood of that country seeking a non-US fuel supplier.

The US Department of Energy (DOE) has committed to acquiring LEU generated from new sources of US domestic enrichment capacity. These can include new enrichment facilities or projects that expand the capacity of existing enrichment facilities. The DOE intends to sell the LEU to utilities operating US reactors to support clean energy generation and sever reliance on Russian imports.

In July 2024 Centrus received a waiver to allow it to import LEU from Russia for delivery to US customers in 2024 and 2025.

In September 2024 Russian President Vladimir Putin said that Russia should consider restricting uranium exports in retaliation for Western sanctions.

Conversion and Enrichment

In March 2023 Cameco and Energoatom signed an agreement to export all of Ukraine's uranium production for processing in Canada. The agreement would see Cameco meet 100% of Energoatom's need for natural uranium hexafluoride from 2024 to 2035 for the country's current nine nuclear reactors.

Western enrichers Orano and Urenco have announced decisions to increase enrichment capacity.

In July 2023 Urenco announced it will increase capacity at its Eunice plant in the USA by 700,000 SWU/yr, setting a target for the first new cascades to be in operation in 2025. In December it announced it would also increase capacity at its Almelo facility in the Netherlands by 750,000 SWU/yr, with first new cascades in operation around 2027.

In October 2023 Orano announced it would increase capacity at its George Besse II facility in France by 2.5 million SWU/yr. Work

began in October 2024, and the first new cascades are to be in operation by 2028. In September 2024 it selected Oak Ridge as the preferred site for a new enrichment facility in the USA.

Fuel fabrication

European countries operating Russian-designed VVER reactors have sought to diversify their fabricated fuel supply. A number of utilities had taken steps to do this prior to February 2022, but developments have accelerated since. VVER-1000 reactors are in operation in Bulgaria and the Czech Republic, and VVER-440 reactors in operation in the Czech Republic, Finland, Hungary and Slovakia.

Czech Republic – following a tender process started in 2018, in April 2022 Czech utility ČEZ selected Westinghouse and Framatome to supply nuclear fuel assemblies for its Temelin nuclear plant. In March 2023 the utility signed an agreement with Westinghouse for supply of fuel for its Dukovany plant. In October 2024 Framatome and ČEZ signed a memorandum of understanding on the development of fuel for VVER-1000 reactors.

Finland – in November 2022 Fortum signed an agreement with Westinghouse for the supply of fuel for its Loviisa plant. The first batch of fuel assemblies supplied by Westinghouse were loaded into unit 2 in September 2024.

Bulgaria – in December 2022 the Kozloduy nuclear power plant signed a 10-year deal with Westinghouse for supply of fuel for unit 5 at the plant. In June 2024 Kozloduy 5 restarted following its annual shutdown with 43 Westinghouse-produced fuel assemblies loaded into the reactor. A similar supply deal was signed with Framatome for unit 6 in January 2023.

Slovakia – in August 2023 Slovak utility Slovenské elektrárne signed an agreement with Westinghouse for the licensing and supply of VVER-440 assemblies to be used in its power plants. In July 2024 a similar agreement was signed with Framatome, with supply of fuel to start from 2027.

Ukraine – in September 2023 Westinghouse fuel was loaded into a VVER-440 reactor for the first time at Ukraine's Rivne plant. The assemblies were supplied under a contract signed in September 2020.

Hungary – in October 2024 a long-term contract was signed with Westinghouse for the supply of fuel from 2027 to the four VVER-

440 reactors in operation at Hungary's Paks nuclear power plant.

Notes & references

General sources

IAEA web portal on the conflict in Ukraine
International Atomic Energy Agency, Nuclear Safety, Security and Safeguards in Ukraine: Summary Report by the Director General, 24 February – 28 April 2022 (28 April 2022)
Situation of nuclear facilities in Ukraine, Institut de radioprotection et de sûreté nucléaire, (1 March 2022)

References

1. WENRA position on the safety situation of Zaporizhzhia NPP after reported shelling activities, Western European Nuclear Regulators Association (10 August 2022) [Back]
2. National Report On Compliance of Ukraine with Obligations under the Convention on Nuclear Safety, State Nuclear Regulatory Inspectorate of Ukraine (June 2019) [Back]
3. Nuclear Safety, Security and Safeguards in Ukraine, 2nd Summary Report by the Director General, International Atomic Energy Agency (6 September 2022) [Back]

Related information

Ukraine
Chernobyl Accident
Security of Nuclear Facilities and Material
Safety of Nuclear Power Reactors